Suraj Swanand Sant

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SUMMARY

CCNA-certified network automation engineer with an MS in Computer Systems Networking and hands-on experience configuring and troubleshooting routers, switches, access points, and Wi-Fi (802.11). Strong in TCP/IP, DNS, DHCP, OSPF, and BGP, with Python and Bash automation for connectivity testing and a root cause approach to reproducing and resolving issues.

EDUCATION

Northeastern University, **Boston, MA, USA –** MS in Computer Systems Networking and Telecommunications **Sep 2023 – Dec 2025**

University of Mumbai, MH, India – Bachelor of Engineering, Electronics and Telecommunication **Aug 2019 – May 2023**

CERTIFICATIONS

- Cisco Certified Network Associate (CCNA) – Issued May 2026 [Link]
- Fortinet NSE 1 / Introduction to the Threat Landscape 3.0 – Issued May 2026 [Link]
- AWS Academy Cloud Architecting Graduate – Issued May 2025 [Link]

EXPERIENCE

The KeelWorks Foundation, Site Reliability Engineer – (Remote) *Oak Harbor, WA, USA* **Apr 2026 – Present**

- Securing production web infrastructure using Cloudflare, configuring WAF rules and SSL/TLS to protect against web-based attacks and enforcing encrypted client-server communication, while provisioning and maintaining Linux server hosts with SSH key authentication and configuring DNS records to keep production systems accessible and secure.
- Monitoring site availability and responding to reliability and security incidents, troubleshooting connectivity and service issues and documenting resolutions and system settings for the team.

Central Lincoln People's Utility District, Telecommunication Engineer Intern – *Newport, OR, USA* **Jul 2024 – Dec 2024**

- Consolidated three SCADA routers into a single Cisco ISR by migrating OSPF, IPAM, and ACLs, cutting hardware footprint by 66%, then verifying the migration by pinging required IPs, confirming OSPF neighbor adjacencies formed, and testing ACLs against restricted IPs.
- Troubleshot a flapping link using PRTG monitoring, analyzing the alert pattern and working down through the physical layer to isolate the root cause (a loose fiber cable at a substation) and restore reliable connectivity.
- Configured, deployed, and tested SEL industrial switches, NTP servers, and SCADA communications across a Layer 3 substation network, completing a hardware refresh of aging devices before rollout.
- Worked with telecom vendors to restore damaged fiber through splicing and Layer 1 optical continuity testing, sustaining sub-second failover and 99.9% availability.

Siemens, Electrical Engineer Intern – *Navi Mumbai, MH, India* **Jun 2022 – Jul 2022**

- Diagnosed and resolved network integration faults in industrial control systems by tracing Modbus and Profibus protocol behavior with analyzers, restoring real-time communication between PLC control units and field devices.
- Validated signal integrity and protocol conformance during commissioning, documenting root-cause findings in handoff reports.

Ericsson, Network Engineer Intern – *Mumbai, MH, India* **Dec 2021 – Jan 2022**

- Implemented DHCP scopes, static routes, and ACL-based traffic filtering across simulated LAN/WAN lab topologies, verifying address allocation and end-to-end reachability to validate connectivity and traffic segmentation.
- Built multi-VLAN enterprise lab topologies in Cisco Packet Tracer using inter-VLAN routing, 802.1Q trunking, and STP, and resolved VLAN tagging and IP subnetting issues through MAC table, ARP, and routing table analysis.

PROJECTS

Two-Site Network Automation Lab (Python, Netmiko, EVE-NG) | [Link]

- Built a two-site enterprise network in EVE-NG with four Layer 3 switches, two edge routers and dual-ISP redundancy, running OSPF internally and eBGP to both ISPs, with PAT and extended ACLs isolating Guest from Staff VLANs, and verified ISP failover and segmentation.
- Automated VLAN, SVI, OSPF, BGP, NAT and ACL configuration with Python and Netmiko, generating per-device configs from structured data and pushing them over an out-of-band management network with per-device error handling; troubleshot SSH algorithm mismatches between legacy IOS and modern clients.

Enterprise Multi-Site Hospital Network Design & Security Implementation (Cisco Packet Tracer) | [Link]

- Architected a multi-site hospital network with VLAN-based segmentation, OSPF dynamic routing, inter-VLAN routing on Layer 3 switches connected to an edge router at each location, and redundant ISP connectivity with automatic failover; designed and deployed wireless access points with WPA2-PSK, configuring dedicated SSIDs and VLANs to separate clinical device, staff, and guest traffic.
- Hardened site-to-site communication with GRE over IPSec tunnels, extended ACLs, PAT, SSH-only management, and port security, and maintained topology diagrams and design documentation in a GitHub repository.

Advanced Network Configuration and Troubleshooting (EVE-NG)

- Engineered a BGP lab in EVE-NG connecting PE, CE, and provider (P) routers, establishing PE-CE relationships and BGP peering to route traffic across a service provider topology.
- Reproduced and resolved connectivity failures across two lab testbeds by correcting ACL misconfigurations and EIGRP-to-OSPF redistribution errors and implemented OSPF area summarization to reduce routing table size.

Linux Network Infrastructure & Security Project (Ubuntu)

- Installed and configured fault-tolerant dual-stack (IPv4/IPv6) DNS and DHCP servers on Ubuntu Linux with primary/secondary failover, sustaining uninterrupted name resolution during simulated outages.
- Secured internal communications with SSL/TLS and site-to-site IPSec VPN tunnels, validated Man-in-the-Middle resistance with arpspoof and Wireshark, and automated backup and restore with PowerShell and Bash via cron.

SKILLS

- **Networking Protocols:** TCP/IP, Ethernet, DNS, DHCP, BGP, OSPF, VLANs, 802.1Q, STP, RSTP, LACP, GRE, HSRP, SNMP, IPv4/IPv6, LAN/WAN.
- **Network Infrastructure:** Cisco IOS, Cisco Catalyst, Cisco ISR, Arista CLI, 802.11 Wi-Fi (SSIDs, WPA2-PSK, 802.1X, RADIUS), NTP servers.
- **Security & Firewalls:** Firewall, ACLs, GRE over IPSec, NAT/PAT, AAA, port security, SSH, SSL/TLS, Cloudflare WAF.
- **Systems & Tooling:** Linux (Ubuntu), VMware, Oracle VirtualBox, EVE-NG, Python, Bash, Netmiko, AI-assisted development (Claude).
- **Monitoring & Documentation:** PRTG, Wireshark, iPerf, Syslog, IPAM, runbooks, root cause analysis, network diagrams, SOPs.